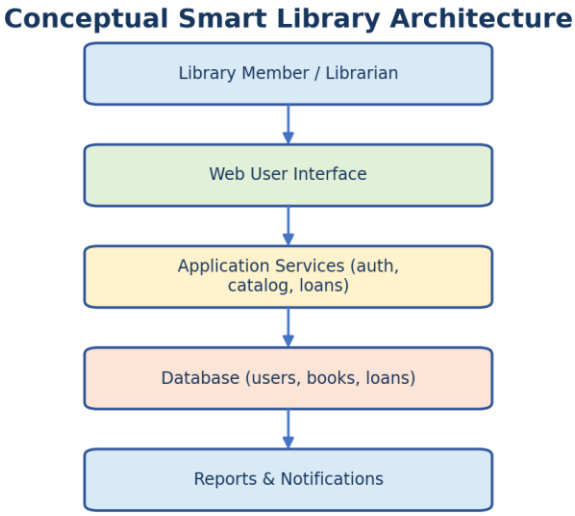
MINI PROJECT REPORT

ENTERPRISE SMART LIBRARY MANAGEMENT PLATFORM



AGILE PLANNING & SCRUM IMPLEMENTATION

Subject: DevOps / Agile Project Planning

Project Type: Mini Project Documentation

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Certificate

This is to certify that the mini project report entitled “Enterprise Smart Library Management Platform – Agile Planning & Scrum Implementation” has been prepared as part of the academic learning activity. The report documents a proposed library management product and demonstrates how Agile concepts and Scrum practices can be applied to plan, organize, review and improve a software project.

The work includes a product vision, stakeholder identification, epics, user stories, acceptance criteria, a prioritized Product Backlog, story-point estimates, a Sprint 1 plan, a Scrum Board and simulations of Scrum events.

Teacher / Guide Signature: _____

Student Signature: _____

Date: _____ Place: _____

Declaration

I declare that this report has been prepared for educational purposes using the project scenario and activity points provided in the trainer guide. The system is presented as a planned software solution; the document does not claim that every feature has already been implemented or deployed.

Acknowledgement

I express my sincere thanks to the subject teacher and institution for providing the mini-project activity on Agile Planning and Scrum. The structured activities helped organize the library platform idea into manageable work, clarify team responsibilities and understand how feedback supports continuous improvement.

Abstract

The Enterprise Smart Library Management Platform is a proposed software system for managing library users, books, borrowing transactions, notifications, reports and administrative activities. Traditional manual or fragmented processes can require repeated data entry, make availability checks slower and create difficulty in maintaining accurate issue and return records. The proposed platform aims to provide a secure and efficient way to manage these activities.

This report focuses not only on the product idea but also on the Agile planning process used to organize its development. A Scrum Team defines the product vision, identifies stakeholders, decomposes the scope into six epics, writes user stories and measurable acceptance criteria, and orders a Product Backlog according to business value, dependencies, risk and stakeholder needs.

The first Sprint is planned around a working foundation for user authentication and basic library access. The plan includes repository setup, Git workflow, project structure, registration, login, a user database, unit testing and documentation. The report also describes the Scrum Board, Sprint Planning, Daily Scrum, Sprint Review and Sprint Retrospective. These practices help the team inspect progress, collect feedback and adapt the backlog for future increments.

Keywords

Agile, Scrum, library management, Product Vision, Epic, User Story, Acceptance Criteria, Product Backlog, Story Points, Sprint Planning, Scrum Board, continuous improvement.

Table of Contents

Chapter	Contents
1	Introduction and Project Objective
2	Scenario, Scope and Expected Outcomes
3	Product Vision and Stakeholders
4	Scrum Team and Responsibilities
5	Epics and Conceptual Architecture
6	User Stories and Acceptance Criteria
7	Product Backlog and Prioritization
8	Story Point Estimation
9	Sprint 1 Plan
10	Scrum Board and Sprint Planning
11	Daily Scrum, Review and Retrospective
12	End-to-End Flow, Deliverables and Conclusion

1. Introduction and Project Objective

1.1 Introduction

A library supports learning and research by organizing books and other learning resources for its members. As collections and membership grow, tasks such as registering members, searching the catalog, issuing books, recording returns and preparing reports become more demanding. A digital platform can centralize these operations and make records easier to maintain.

The Enterprise Smart Library Management Platform is planned as a single system for librarians, members, administrators and management. It includes account access, catalog maintenance, borrowing workflows, reminders, reports and administrative controls. The project is intentionally planned in increments so that stakeholders can review the product after each Sprint.

1.2 Project Objective

The primary objective is to enable students to apply Agile concepts by collaboratively planning a complete software project. The team transforms a product idea into a Product Vision, epics, user stories, acceptance criteria, Product Backlog and Sprint 1 plan, then simulates Scrum events to see how inspection, feedback and adaptation improve the product.

1.3 Expected Learning Outcomes

- Explain how Agile concepts apply to a realistic software project.
- Collaboratively define a Product Vision and identify stakeholders.
- Decompose product scope into Epics and User Stories.
- Write measurable Acceptance Criteria.
- Prioritize and estimate a Product Backlog.
- Create and use a Scrum Board.
- Plan and simulate a Sprint.
- Participate effectively in Daily Scrum, Sprint Review and Sprint Retrospective.
- Explain how the product could later be scaled using LeSS, SAFe or Scrum@Scale.

1.4 Project Approach

The team starts by clarifying value and user needs. It then organizes requirements into a backlog, selects achievable work for a Sprint, develops and tests an increment, demonstrates the outcome to stakeholders and records improvement actions. The next Sprint uses the feedback and updated priorities.

2. Project Scenario and Scope

The project scenario describes a software system that manages library users, books, borrowing, notifications, reports and administration. Students act as an Agile delivery team. The goal is to plan a product that can evolve through feedback instead of attempting to deliver every function at once.

2.1 Functional Scope

Epic	Included Functions
User Management	Registration, login, logout, password management and user profiles.
Book Management	Add, update, delete and search books; maintain book categories.
Borrowing Management	Issue, return and renew books; manage due dates and calculate fines.
Notifications	Due-date reminders, overdue notifications and new-book alerts.
Reporting	Books issued, overdue books, popular books and user activity.
Administration	Role management, backup, audit logs and system configuration.

2.2 Scope Boundaries

The six epics define the planned product scope. Sprint 1 is deliberately smaller than the full product and concentrates on registration, login, the user database and supporting development practices. Advanced reports, notification rules and administrative features can be scheduled after the foundation is validated.

2.3 Why Incremental Delivery Matters

- Stakeholders can review working functionality early.
- The team can discover unclear requirements before too much work is completed.
- High-value features can be delivered before lower-priority enhancements.
- Testing and feedback can influence the next Sprint plan.

3. Product Vision and Business Value

3.1 Sample Product Vision

"For librarians, library members and administrators, the Enterprise Smart Library Management Platform provides a secure and efficient way to manage users, books, borrowing, notifications and reports, reducing manual work and improving the library experience."

3.2 Target Users

Target user	Primary need
Librarians	Register members, manage the catalog and record issue/return operations.
Library members	Search for books, check borrowing status and receive notifications.
Library administrators	Configure the system, manage roles, view reports and audit activity.
Management	Understand usage, performance and business/library activity.
IT / Support team	Support deployment, availability, security and maintenance.

3.3 Main Problem

The product addresses the need to coordinate multiple library activities in a reliable way. When member information, catalog details and borrowing records are maintained through disconnected or manual processes, staff may need to repeat work and members may find it difficult to know whether a book is available.

3.4 Expected Value

- Reduce repetitive manual work for library staff.
- Improve the consistency of member and borrowing records.
- Make catalog search and book availability information easier to access.
- Support timely returns through reminders and overdue notifications.
- Help management understand usage through reports and statistics.
- Provide administrative controls for roles, backups and audit logs.

3.5 Vision Quality Check

A useful Product Vision names the users, identifies the problem and expresses the value created. It should remain concise and outcome-focused rather than becoming a long list of technical implementation details.

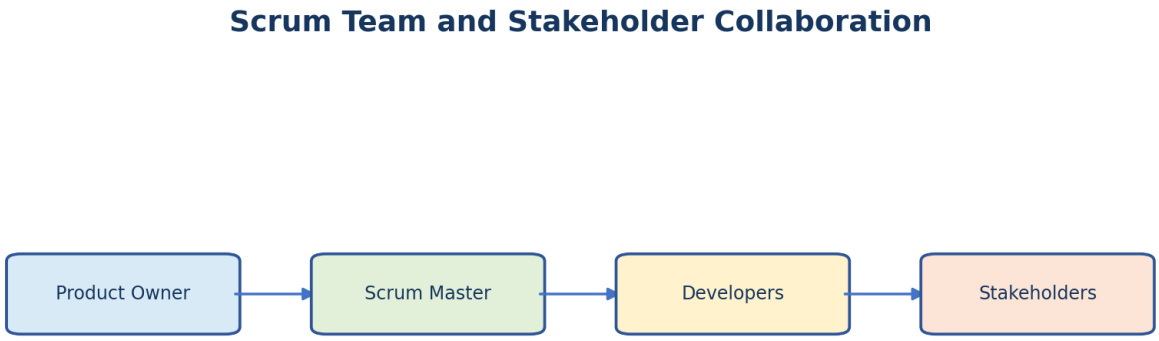
4. Stakeholders and Scrum Team

4.1 Stakeholder Analysis

Stakeholder	Interest / Need	How feedback helps
Library Administrator	Configuration, reports, audit and administration	Confirms access rules and operational controls.
Librarian	Registration, catalog management, issue/return	Validates everyday workflows.
Library Member	Search, borrowing status and notifications	Checks ease of finding and borrowing items.
Management	Usage, performance and business reports	Guides reporting priorities.
IT / Support Team	Deployment, availability, security and maintenance	Identifies operational and technical risks.

4.2 Scrum Team Roles

Role	Responsibilities
Product Owner	Defines product value, clarifies requirements, orders the Product Backlog and accepts or rejects work against agreed criteria.
Scrum Master	Facilitates Scrum events, helps remove impediments, supports Scrum understanding and encourages continuous improvement.
Developers	Design, implement, test, document and integrate the product increment. The group jointly decides how to complete Sprint work.
Stakeholders	Represent users and other interested parties and provide needs, questions and feedback.



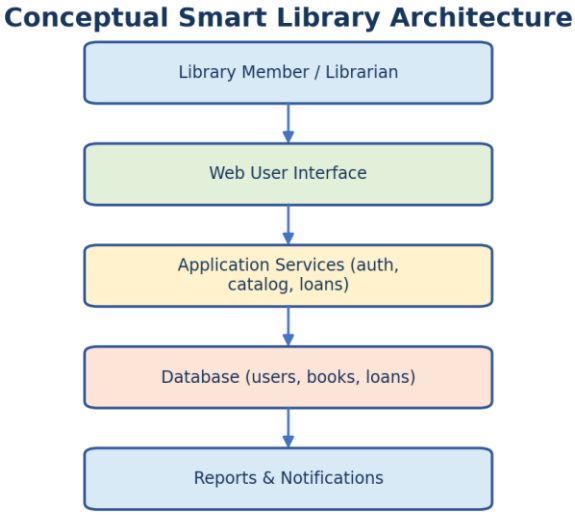
For the classroom activity, a team of approximately 5–8 students is recommended. One student can act as Product Owner, one as Scrum Master and the remaining members as Developers. The Developers collectively own the plan for completing the selected work.

5. Epics and Conceptual Architecture

An Epic is a large functional area that can be decomposed into multiple User Stories. The six epics below organize the system at a high level and provide a structure for backlog refinement.

Epic	Example stories / capabilities
User Management	Register member; log in and out; reset password; maintain profile.
Book Management	Add a book; update metadata; delete or deactivate a record; search catalog; manage categories.
Borrowing Management	Issue book; return book; renew loan; calculate due date; calculate fine.
Notifications	Send due-date reminder; notify overdue item; announce new books.
Reporting	View issued books; list overdue books; identify popular books; review user activity.
Administration	Assign roles; back up data; review audit logs; configure system.

5.1 Conceptual Architecture



The diagram is a conceptual architecture for planning and discussion. It shows a user-facing interface connected to application services and a database, with reports and notifications supported by the system. It is not a detailed implementation specification, and the guide does not require every layer to be implemented during Sprint 1.

5.2 Architectural Considerations

- Separate the user interface from core application rules where practical.
- Store users, books and borrowing records consistently.
- Protect functions according to user roles.
- Plan backup and audit features as part of administration.
- Keep the first Sprint focused on a small working foundation.

6. User Stories

A User Story describes a capability from the perspective of a user and explains why that capability matters. The standard format is: "As a [User], I want [Functionality], so that [Business Value]." Stories help the team discuss user outcomes rather than only technical tasks.

Story	User Story
Register New User	As a librarian, I want to register new library members, so that they can borrow books.
Search Books	As a library member, I want to search books by title, so that I can quickly locate available books.
Issue Book	As a librarian, I want to issue books to registered members, so that borrowing records are maintained automatically.
Return Book	As a librarian, I want to record returned books, so that book availability and borrowing history remain accurate.
Due Date Reminder	As a library member, I want to receive a due-date reminder, so that I can return books on time.

6.1 Additional Catalog Stories from the Guide

Barcode-assisted cataloging: As a librarian, I want to scan an ISBN or barcode to auto-populate book metadata such as title, author, publisher and genre, so that I can catalog new acquisitions in seconds.

Catalog filtering: As a patron, I want to filter the online catalog by availability, author and format (eBook or hardcopy), so that I can quickly locate items ready for pickup.

6.2 Story-Writing Checklist

- Identify a specific user or stakeholder.
- Describe one clear function or outcome.
- State the reason or value to the user.
- Keep the story small enough to discuss, estimate and test.
- Add acceptance criteria to clarify what successful completion means.

7. Acceptance Criteria

Acceptance Criteria are the conditions that must be satisfied before a User Story can be considered complete. They reduce ambiguity and give Developers, the Product Owner and stakeholders a shared understanding of the expected result.

7.1 Example: Issue Book

- The book must be available.
- The user must be registered.
- The due date should be generated automatically.
- The borrowing history should be updated.
- A confirmation message should be displayed.

7.2 Barcode / ISBN Metadata

For the barcode-assisted cataloging story, metadata is fetched through an open book API and the returned fields remain manually editable. The librarian should be able to correct data where the external result is incomplete or inaccurate.

7.3 Search Filters and Performance

For the catalog-filtering story, search results should update in under one second, and unavailable books should show expected return dates. These criteria are measurable and can be checked during testing.

7.4 Acceptance Criteria Table

Story	Acceptance checks
Register New User	Required member details are validated; a valid member record is saved; duplicate or invalid input is handled clearly; confirmation is shown.
Search Books	Search by title returns relevant results; availability is visible; no-match searches show a clear message.
Issue Book	Book is available; member is registered; due date is generated; loan history updates; confirmation is displayed.
Return Book	Return is recorded; availability is updated; borrowing history remains accurate.
Due Date Reminder	Reminder is associated with the correct loan and due date; delivery outcome can be checked.

The registration and reminder checks in this table are suggested elaborations for planning. The specific criteria explicitly supplied in the guide are reproduced in the relevant examples above.

8. Product Backlog and Prioritization

The Product Backlog is an ordered, evolving list of work needed to improve the product. The Product Owner orders items by business value, dependencies, risk and stakeholder needs. It remains a living list and may change after feedback.

Order	Backlog item	Priority	Business value
1	User Registration	High	Essential
2	User Login	High	Essential
3	Search Book	High	Essential
4	Add Book	High	Essential
5	Issue Book	High	Essential
6	Return Book	High	Essential
7	Password Reset	High	High
8	Due Date Reminder	Medium	Medium
9	Daily Library Report	Medium	Medium
10	Borrowing Statistics	Medium	Medium
11	Analytics Dashboard	Low	Low
12	Audit Logs	Low	Low

8.1 Why Prioritization Matters

- High-priority items establish basic access and essential library operations.
- Dependencies help the team choose a sensible implementation order.
- Risky or uncertain work may need early discussion or investigation.
- Stakeholder feedback can change the relative importance of backlog items.

8.2 Backlog Refinement

During refinement, the team clarifies the purpose of an item, checks acceptance criteria, discusses dependencies and estimates effort. Items that are too large can be split into smaller stories. The Product Owner remains responsible for ordering the backlog, while Developers contribute technical understanding.

9. Story Point Estimation

The guide recommends a Fibonacci-like scale such as 1, 2, 3, 5, 8 and 13. Story Points express relative effort, complexity and uncertainty. They are not exact hours and should not be converted mechanically into time.

Story	Sample Story Points	Interpretation for discussion
Register User	3	A small feature with validation and data persistence.
User Login	2	A relatively small authentication function.
Search Book	3	A basic search and results flow.
Add Book	5	A form and save flow with more fields and validation.
Issue Book	8	A more complex workflow with availability, due date and history updates.
Return Book	5	A transaction that updates availability and history.
Password Reset	5	A flow involving identity checks and password update.

9.1 Planning Poker

1. The Product Owner explains the story and its acceptance criteria.
2. Each Developer selects an estimate independently.
3. Team members reveal estimates together.
4. The team discusses major differences and hidden assumptions.
5. The team estimates again until it reaches a useful shared view.

9.2 Estimation Principles

- Estimate the work relative to other stories, not in hours.
- Consider complexity, effort and uncertainty.
- Clarify acceptance criteria before estimating.
- Use discussion to uncover risks and dependencies.
- Treat the values as planning aids, not promises.

10. Sprint 1 Plan

10.1 Sprint Goal

"Deliver a working foundation for user authentication and basic library access."

Sprint 1 focuses on foundational technical work and the first usable user-facing functions. The team should select a realistic amount of work based on capacity and dependencies.

Sprint Backlog item	Suggested points	Purpose
Create repository	2	Provide a shared place for source code and project history.
Configure Git workflow	2	Agree on a basic way to collaborate on changes.
Project structure	3	Organize the application files and modules.
User Registration	3	Allow creation of a library user record.
User Login	2	Allow an existing user to access the system.
User Database	5	Store the user information required by authentication.
Unit Testing	3	Check implemented functions in a repeatable way.
Documentation	2	Record setup and usage information.

10.2 Sprint 1 Deliverables

- Working registration module.
- Working login module.
- Basic user database.
- Automated/unit tests for implemented functions.
- Updated repository and documentation.
- Sprint Review demonstration.

10.3 Definition of Done

The guide asks the team to agree on a Definition of Done. For this plan, a practical draft is: code is integrated into the shared repository, the selected acceptance criteria are met, relevant tests pass, known limitations are recorded and setup instructions are updated. The team should agree on the exact wording before work begins.

11. Scrum Board and Work Visibility

A Scrum Board visualizes the state of selected Sprint work. The guide gives the sequence: To Do → In Progress → Code Review → Testing → Done. Team members move a card only when the work actually changes state.

Example Scrum Board



11.1 Example Board Use

Column	Meaning	Example
To Do	Selected work not yet started	Unit testing task waiting to begin.
In Progress	Someone is actively working on it	Registration form under development.
Code Review	Implementation is ready for review	Login code awaiting teammate review.
Testing	Function is being checked against expected behavior	Registration validation tests running.
Done	Work meets the agreed Definition of Done	Repository setup completed and documented.

11.2 Board Working Agreements

- Keep cards clear and small enough to track.
- Move cards only when their state genuinely changes.
- Make blocked items visible and ask for help promptly.
- Avoid too much simultaneous work in progress.
- Use the board to coordinate toward the Sprint Goal, not as a substitute for collaboration.

12. Sprint Planning

Sprint Planning establishes why the Sprint is valuable, what can be completed and how the selected work will be done. The team should use the Product Goal, Sprint Goal, ordered backlog, team capacity and technical dependencies to make a realistic plan.

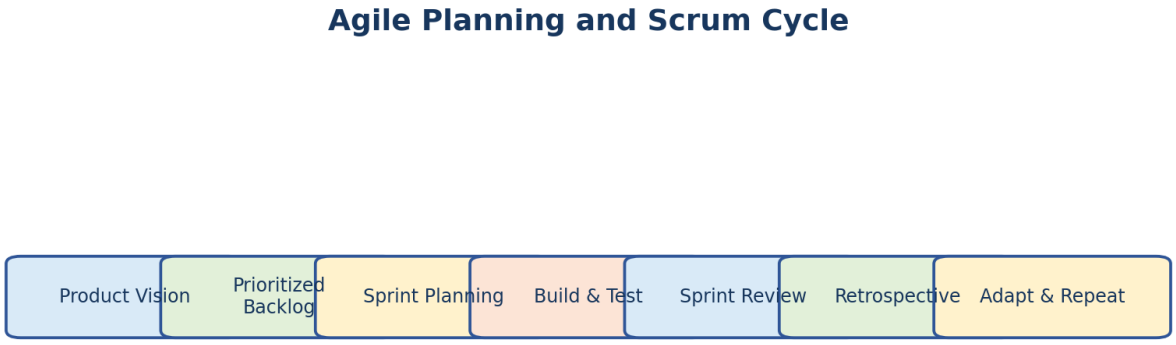
6. Confirm the Product Goal and Sprint Goal.
7. Review the highest-priority Product Backlog items.
8. Clarify User Stories and Acceptance Criteria.
9. Estimate or confirm Story Points.
10. Discuss dependencies, risks and technical work.
11. Select a realistic amount of work based on team capacity.
12. Break selected stories into implementable tasks.
13. Agree on the Definition of Done.
14. Create the Sprint Backlog and Scrum Board.

12.1 Planning Questions

- Does the selected work support the Sprint Goal?
- Are requirements clear enough to start?
- Does the team understand the dependencies?
- Is the workload realistic for the available team capacity?
- How will the team test and demonstrate each completed item?

12.2 Planning Output

The main output is a Sprint Backlog that contains the Sprint Goal, selected backlog items and a workable plan for delivering them. During the Sprint, Developers can adjust the plan as they learn more, while keeping focus on the Sprint Goal.



13. Daily Scrum Simulation

The Daily Scrum is a short coordination event for Developers to inspect progress toward the Sprint Goal and adapt their plan. The classroom simulation should focus on progress, next steps and impediments rather than becoming a status report only for the Scrum Master.

13.1 Three Prompt Questions

- What did I complete?
- What will I work on next?
- What impediment or dependency needs attention?

13.2 Sample Daily Scrum Notes

Team member / area	Completed	Next	Impediment / action
Registration developer	Basic form layout	Connect validation and save operation	Needs agreed required fields; ask Product Owner.
Authentication developer	Login flow outlined	Connect login to user database	Waiting for database schema discussion.
Database developer	User table drafted	Test create/read operations	Confirm unique member identifier.
Testing / documentation	Test checklist drafted	Write tests for registration	Needs stable interface and expected messages.

The notes above are an illustrative classroom example, not evidence that implementation work has already been completed. A real team should record its actual progress and blockers.

13.3 Expected Benefits

- Expose blockers early.
- Coordinate work between Developers.
- Reduce duplicate effort and conflicting changes.
- Keep the team focused on the Sprint Goal.

14. Sprint Review

At the end of the simulated Sprint, the team demonstrates the working increment and discusses the outcome with stakeholders. The purpose is to inspect the product and decide what should happen next, not merely to present slides.

14.1 Review Activities

- Demonstrate completed functionality.
- Show how acceptance criteria are satisfied.
- Collect stakeholder feedback.
- Identify unfinished work and known limitations.
- Update or reorder the Product Backlog.

14.2 Example Review Scenario

The team demonstrates User Registration and Login. The librarian stakeholder asks for a member ID field, and the administrator asks for a role-based access option. The Product Owner records these as new or refined Product Backlog items for future prioritization.

14.3 Sample Review Record

Item	Example record
Increment demonstrated	Registration and login foundation.
Stakeholder feedback	Add a member ID field; consider role-based access.
Acceptance criteria status	Record which criteria passed and which remain incomplete.
Unfinished work	List any incomplete or blocked items honestly.
Backlog action	Add or refine stories and reorder based on value and dependencies.

14.4 Why Feedback Matters

Feedback may reveal missing fields, access needs or usability problems that were not obvious during planning. By recording feedback in the backlog, the team can make evidence-informed decisions about the next increment.

15. Sprint Retrospective

The Sprint Retrospective is an opportunity for the Scrum Team to examine how the Sprint went in terms of people, collaboration, process and tools, and to identify improvements. It should result in a small number of actionable changes.

15.1 Reflection Questions

- What went well?
- What did not go well?
- What caused the difficulties?
- What should we continue?
- What should we stop or change?
- What one or two improvement actions will be applied in the next Sprint?

15.2 Example Improvement Action List

Observation	Improvement action	Owner	Follow-up measure
Requirements were unclear	Review acceptance criteria before estimating	Product Owner + Developers	Stories selected for next Sprint have clear criteria.
Database and login work were dependent	Agree on schema early and share it in repository	Database / authentication Developers	Integration blocker is checked at next Daily Scrum.
Some tasks stayed in progress too long	Split tasks and limit work in progress	Whole team	Board shows fewer oversized active tasks.

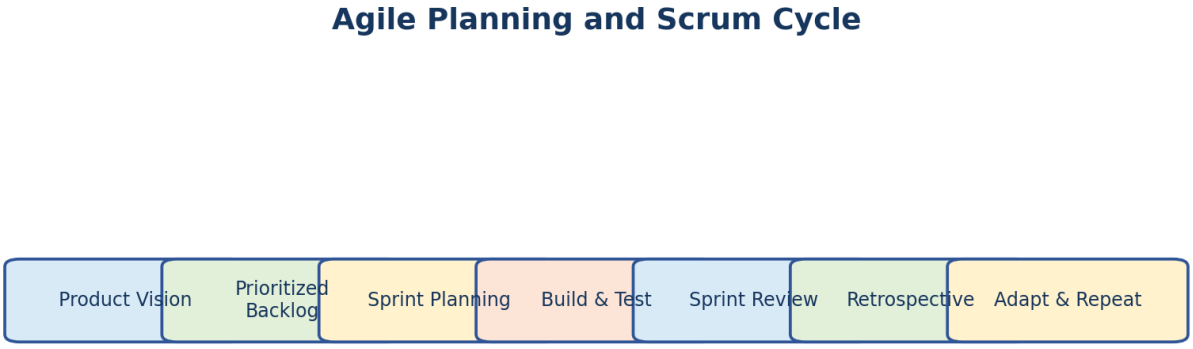
These are sample actions to illustrate the expected output. The actual team should record observations from its own Sprint and assign an owner and measurable follow-up.

15.3 Continuous Improvement

A retrospective is useful only when the team follows through. At the next Sprint, the team checks whether the agreed action helped and adjusts again if necessary. This creates a repeating cycle of learning.

16. End-to-End Project Flow and Scaling

The project follows an end-to-end loop: clarify value → prioritize → plan → build → inspect → receive feedback → adapt → plan the next increment.



16.1 How the Activities Connect

Stage	Output	How it supports the next stage
Product Vision	Clear intended value	Guides stakeholder and backlog decisions.
Stakeholders	Needs and feedback sources	Helps define epics and stories.
Epics / User Stories	Organized product requirements	Makes work discussable and testable.
Acceptance Criteria	Conditions for success	Supports estimation, development and review.
Prioritized Backlog	Ordered work	Guides Sprint selection.
Story Points	Relative effort estimates	Supports capacity-aware planning.
Sprint Backlog / Board	Selected work and visible status	Coordinates development and testing.
Review / Retrospective	Product feedback and process actions	Improves the next increment and the way of working.

16.2 Future Scaling Options

The trainer guide asks students to explain how the product could later be scaled using LeSS, SAFe or Scrum@Scale. These are possible approaches for coordinating work when several teams contribute to a larger product. At the current mini-project stage, one Scrum Team is sufficient; the guide does not require a scaling framework to be implemented.

- LeSS: an approach for applying Scrum with multiple teams working on one product.
- SAFe: a framework for coordinating Agile work across larger organizations.
- Scrum@Scale: an approach for scaling Scrum through interconnected Scrum structures.

17. Student Deliverables, Conclusion and References

17.1 Student Submission Checklist

- Product Vision.
- Stakeholder Map.
- Epic List.
- User Story Repository.
- Acceptance Criteria.
- Prioritized Product Backlog.
- Story Point Estimation Sheet.
- Sprint Goal and Sprint Backlog.
- Scrum Board screenshot/export.
- Daily Scrum notes.
- Sprint Review report.
- Sprint Retrospective report.
- Architecture diagram.
- Final team presentation.

17.2 Conclusion

The Enterprise Smart Library Management Platform provides a realistic scenario for learning Agile planning and Scrum. Its scope covers user management, book management, borrowing, notifications, reporting and administration. By organizing the product into epics and user stories, the team can describe requirements in user-focused language and define acceptance criteria that make completion clearer.

Prioritization and story-point estimation help the team select valuable work, while Sprint Planning creates a focused and achievable Sprint Goal. The Scrum Board makes work visible, the Daily Scrum supports coordination, the Sprint Review gathers product feedback and the Sprint Retrospective identifies process improvements. Repeating this cycle helps the product evolve incrementally.

The planned Sprint 1 foundation—repository, Git workflow, project structure, registration, login, user database, testing and documentation—provides a starting point for later work. The project also establishes a practical foundation for future DevOps topics such as Git, CI/CD, testing, containers, infrastructure automation and monitoring.

17.3 References / Basis

Primary source: “MINI PROJECT: AGILE PLANNING & SCRUM IMPLEMENTATION — Enterprise Smart Library Management Platform, Trainer Guide + Student Activity Workbook” supplied with the assignment. The report follows the scenario, terminology, activity sequence, sample stories, backlog order, sample points, Sprint 1 plan and submission checklist from that guide.

Note: The architecture and Scrum-cycle graphics are conceptual illustrations for this report. Sample meeting notes and improvement actions are examples for classroom use and should be replaced with actual team observations where required.